MIL-H-14705 (MU)
3 May 1968
SUPERSEDING
PA-PD-2486 (Rev. C)
18 May 1966

MILITARY SPECIFICATION

HANDLING DEVICE, PROJECTILE: H4220 AND H4224 (TRAINING)

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for one type of handling device used to handle the XM454 Projectile and XM455 Training Projectile as applicable and designated as Handling Device, Projectile: H4220 or Handling Device, Projectile: H4224 (Training).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.

MILITARY

MIL-A-46050 - Adhesive, Special Rapid Room Temperature Curing Solventless.
MIL-S-22473 - Sealing, Locking and Retaining Compounds, Single Component.
MIL-W-8604 - Welding of Aluminum Alloys, Process for.

FSC-1190

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:36Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.
MS 15795 - Washer, Flat.
MS 35223 - Screw, Machine, Pan Head, Slotted.
MS 20426 - Rivet, Countersunk.

DRAWINGS

ORDNANCE CORPS

8865534 - Handling Device, Projectile: H4220.
8865608 - Hinge, Cloth.
8865609 - Handle.
8866670 - Handle (Modified).
8866671 - Latch Assembly.
8866672 - Latch Assembly (Modified).
8866673 - Hook.
8866674 - Hook (Modified).
8866675 - Cover Assembly (Left Hand).
8866676 - Cover Assembly (Right Hand).
8866677 - Support.
8866678 - Cover.
8880457 - Handling Device, Projectile: H4224 (Training).
9206906 - Carton, Packing, Ammunition, for Handling Device, Projectile: H4220 or H4224.
9215028 - Cushion.
ET-8865534 - Equipment Tabulation (H4220).
ET-8880457 - Equipment Tabulation (H4224).
AA-38-409 - Dummy Test Projectile T5158.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. - The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specifications listed on the applicable drawing.

3.2 Assembly. - The handling device shall comply with all requirements specified on Drawing (dwgs) 8865534 and 8880457, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

Source: https://assist.dla.mil -- Downloaded: 202640-18T04:36Z
Check the source to verify that this is the current version before use.

3.3 First article testing. - Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Functional test. - Each protective handling device shall fit on a dummy projectile (see dwg AA-38-409) and shall not shift or rotate or show any evidence of scratching dummy projectile when tested as specified in paragraph 4.4.2.

3.5 Dead weight test. - Each handling device shall be tested with a dummy test slug. The handling device shall be placed on the test slug and shall show no evidence of deformation, separation from the covers of the handles, hinge, and rivets nor deterioration of welds when tested as specified in paragraph 4.4.3.

3.6 Workmanship.

3.6.1 Plastic and metal characteristics. - All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.6.2 Threads. - Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.6.3 Burrs. - Parts shall be free from burrs.

3.6.4 Foreign matter. - Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.6.5 Cleaning. - The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. - When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8865534 or 8880457), 3 subassemblies (see dwgs 8866675 and 8866676 and 3 sets of parts, for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. - The first article inspection sample shall be subjected to the inspections and tests specified in table I. Tests need not be conducted in order specified in table I.

Source: https://assist.dla.mil -- Downloaded: 202640-18T04:36Z
Check the source to verify that this is the current version before use.

Table I

First Article Inspection and Tests

Examination or Test	Contractor's Inspection before Submission	Non-destructive	Assemblies	Sub-Assemblies	Parts
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X	X
2. Functional (see 3.4)	X	X	X		
3. Dead Weight (see 3.5)	X	X	X		
4. Tensile (see 4.3.3.1)	X	X			X
5. Cushion bond (see 4.3.3.3)	X	X	X	X	
6. Latches (see 4.3.3.4)	X	X	X		
7. Paint Adhesion (see 4.3.3.5)	X	X	X		

4.2.3 Inspection equipment. - The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.

4.2.4 Rejection. - If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Product acceptance inspection.

4.3.1 Lot formation. - A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.1 Inspection lot. - An inspection lot shall be the quantity of parts, sub-assemblies, or assemblies from the same lot which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. - Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:36Z
Check the source to verify that this is the current version before use.

Tables for Handling Device: H4220

Table II

Major Characteristic

Lot Size	Sample Size
1 - 155	100% inspection
156 - 200	155
201 - 250	175
251 - 300	190
301 - 350	200
351 - 400	210
401 - 450	220
451 - 500	225
501 - 550	230
551 - 600	235
601 - 650	240
651 - 700	243
701 - 750	247
751 - 800	250
801 - 900	254
901 - 1000	258
1001 - 1100	261
1101 - 1200	264
1201 - 1300	266
1301 - 1400	268
1401 - 1500	270
1501 - 1600	272

Table III

Minor Characteristic

Lot Size	Sample Size
1 - 25	100% inspection
26 - 30	25
31 - 35	26
36 - 40	27
41 - 45	28
46 - 50	29
51 - 55	30
56 - 60	31
61 - 65	32
66 - 70	33
71 - 75	34
76 - 85	35
86 - 95	36
96 - 110	37
111 - 130	38
131 - 156	39
157 - 190	40
191 - 240	41
241 - 340	42
341 - 520	43
521 - 1120	44
1121 and over	45

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:36Z
Check the source to verify that this is the current version before use.

Tables for Handling Device: H4224

Table IV

Major Characteristic

Lot Size	Sample Size
1 - 25	100% inspection
26 - 30	25
31 - 35	26
36 - 40	27
41 - 45	28
46 - 50	29
51 - 55	30
56 - 60	31
61 - 65	32
66 - 70	33
71 - 75	34
76 - 85	35
86 - 95	36
96 - 110	37
111 - 130	38
131 - 156	39
157 - 190	40
191 - 240	41
241 - 340	42
341 - 520	43
521 - 1120	44
1121 and over	45

Table V

Minor Characteristic

Lot Size	Sample Size
1 - 25	100% inspection
26 - 35	16
36 - 45	17
46 - 55	18
56 - 65	19
66 - 100	20
101 - 200	21
210 and over	22

4.3.2.1 Classification of characteristics. - Critical, Major, and Minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.01 Hinge, cloth (see dwg 8865608 covering a detail of dwgs 8880457 and 8865534 as applicable).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101.	Diameter of holes, max	Gage
102.	Length, max	Gage
103.	Width, max	Gage
104.	No ragged holes	Visual
105.	No tears in webbing	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:36Z
Check the source to verify that this is the current version before use.

Minor:

201. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.02 Handle (see dwg 8865609 covering a detail of dwg 8866670).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Objective evidence (see 6.2) available for part in accordance with dwg 8865609.....Visual

Minor:

201. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.03 Handle (modified) (see dwg 8866670 covering a detail of dwgs 8866676 and 8866675 as applicable).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Angle of bends.....Gage
102. Protective finish not damaged.....Visual

Minor:

201. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.04 Latch assembly (see dwg 8866671 covering a detail of dwg 8866672).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Objective evidence (see 6.2) available for part in accordance with dwg 8866671.....Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:36Z
Check the source to verify that this is the current version before use.

Minor:

201. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.05 Latch assembly (modified) (see dwg 8866672 covering a detail of dwgs 8880457 and 8865534 as applicable).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Diameter of holes, max	Gage
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

4.3.2.1.06 Hook (see dwg 8866673 covering a detail of dwg 8866674).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) available for part in accordance with dwg 8866673	Visual
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

4.3.2.1.07 Hook (modified) (see dwg 8866674 covering a detail of dwgs 8880457 and 8865534 as applicable).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Diameter of holes, max	Gage
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

Source: https://assist.dla.mil -- Downloaded: 20261O-18T04:36Z
Check the source to verify that this is the current version before use.

4.3.2.1.08 Support (see dwg 8866677 covering a detail of dwgs 8866676 and 8866675 as applicable).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Pitch diameter of threaded holes	Gage
102.	Minor diameter of threaded holes, max	Gage
103.	Radius of curved surface	Gage
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

4.3.2.1.09 Cover (see dwg 8866678 covering a detail of dwgs 8866675 and 8866676 as applicable).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Diameter of holes, max	Gage
102.	Diameter of countersink, min	Gage
103.	Edges rounded	Visual/Manual
Minor:		
201.	Workmanship (see 3.6)	Visual/Manual

4.3.2.1.10 Cover assemblies left and right hand (see dwgs 8866675 and 8866676 covering a detail of dwgs 8880457 and 8865534 as applicable).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Location of supports	Gage
102.	Welding in accordance with MIL-W-8604	Visual
103.	Handle fastened securely	Manual
104.	Objective evidence (see 6.2) available for protective finish in accordance with dwgs 8866675 and 8866676 as applicable	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:36Z
Check the source to verify that this is the current version before use.

Minor:

201. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.11 Cushion (see dwg 9215028 covering a detail of dwgs 8880457 and 8865534).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Objective evidence (see 6.2) available for part in accordance with dwg 9215028.....Visual

Minor:

201. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.12 Washer, flat (see dwg MS15795-206); Screw, machine, pan head, slotted (see dwg MS35223-27); rivet, countersink (see dwg MS-20426A4-7) covering details of dwgs 8865534 and 8880457 as applicable.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Objective evidence (see 6.2) available for part in accordance with applicable standard.....Visual

Minor: None defined.

4.3.2.1.13 Handling device, projectile (see dwgs 8865534 and 8880457 as applicable) before painting.

Categories	Characteristics	Method of inspection

Critical:

1. Foreign material not bonded to rubber cushion.....Visual

Major:

101. Objective evidence (see 6.2) available for adhesive in accordance with MIL-A-46050.....Visual
102. Objective evidence (see 6.2) available for sealing and retaining compound in accordance with MIL-S-22473.....Visual

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:36Z
Check the source to verify that this is the currentversion before use.

103. Rubber does not overlap metal more than dimension specified on dwg.....Gage
104. Latches operate properly (see 4.3.3.4).....Manual
105. No component loose.....Manual
106. No component missing.....Visual
107. Objective evidence (see 6.2) available for protective finish in accordance with dwgs as applicable.....Visual

Minor:

201. Marking.....Visual
202. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.14 Handling device, projectile (see dwgs 8865534 and 8880457 as applicable).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major: None defined.

Minor:

201. Paint present no bare spot.....Visual
202. Workmanship (see 3.6).....Visual/Manual

4.3.2.1.15 Carton, packing and section 5 ammunition, for handling device, projectile; H4220 or H4224 (see dwg 9206906) before closing cover.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:

101. Sealing strip present on bottom of carton.....Visual
102. Item wrapped and secured with tape.....Visual
103. Item tight in carton.....Manual

Minor:

201. Workmanship (see 3.6).....Visual/Manual

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:36Z
Check the source to verify that this is the current version before use.

4.3.2.1.16 Carton, packing, ammunition, for handling device, projectile: H4220 or H4224 (see dwg 9206906 and section 5).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
	101. Carton sealed properly	Visual/Manual
	102. Label waterproofed	Visual
Minor:		
	201. Marking	Visual
	202. Workmanship (see 3.6)	Visual/Manual

4.3.3 Testing. - The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

Test	Test Classification
a. Tensile	Major
b. Functional	Major
c. Dead Weight	Major
d. Cushion Bond	Major
e. Latches	Major
f. Paint Adhesion	Minor

None of these tests are considered destructive and samples so tested may be returned to the lot.

4.3.3.1 Tensile test (see dwgs 8866671 and 8866673). - A sample of latch assemblies or hooks shall be selected at random from each lot and tested as specified in 4.4.1. If any latch assembly or hook fails to comply with the drawing requirements, the lot of parts shall be rejected.

4.3.3.2 Functional (see 3.4) - Dead weight (see 3.5). - Each projectile handling device shall be subjected to these tests (see paragraph 4.4.2 or 4.4.3 as applicable. If any protective handling device fails to comply with 3.4 or 3.5 as applicable, it shall be classified defective and removed from the lot.

4.3.3.3 Cushion Bond (see dwgs 8865534 and 8880457). - The bond of the cushions to the handling device shall be tested as specified in 4.4.4 for each assembly in the lot. Any indication of the cushion not being bonded to the metal shall reject the assembly. Defective assemblies shall be removed from the lot.

4.3.3.4 Latches (see dwgs 8865534 and 8880457). - Each latch on each assembly in the lot shall be tested as specified in 4.4.5. Any latch which does not function properly with normal thumb pressure shall reject the assembly. Defective assemblies shall be removed from the lot.

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:36Z
Check the source to verify that this is the current version before use.

4.3.3.5 Paint adhesion (see dwgs 8865534 and 8880457). - Each part or assembly shall be tested for paint adhesion as specified in 4.2.6 TT-C-490. Any part or assembly which fails to comply with 3.7 TT-C-490 shall be classified defective and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. - Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8865534 or 8880457 as applicable. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. - Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by a Government representative authorized by the procuring agency.

4.4 Test methods. - The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Tensile test (see dwgs 8866671 and 8866673). - A sample shall be selected at random from each inspection lot in accordance with table VI. Each sample latch or hook shall be pulled on approved equipment to the minimum pounds specified as working strength on the applicable drawing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:36Z
Check the source to verify that this is the current version before use.

Table No. VI

Sample Sizes According to Lot Size

Lot Size	Sample Size for No Failures 100% Inspection
1 - 25	25
26 - 30	26
31 - 35	27
36 - 40	28
41 - 45	29
46 - 50	30
51 - 55	31
56 - 60	32
61 - 65	33
66 - 70	34
71 - 75	35
76 - 85	36
86 - 95	37
96 - 110	38
111 - 130	39
131 - 156	40
157 - 190	41
191 - 240	42
241 - 340	43
341 - 520	44
521 - 1120	45
1121 and over	

4.4.2 Functional test (see 3.4). - The handling device shall be placed on a dummy projectile (obtained from the contracting officer in accordance with dwg AA-38-409), lifted horizontal to the ground and then rotated ninety (90) degrees to the vertical plane.

4.4.3 Dead weight test (see 3.5). - The handling device shall be tested with a steel dummy test slug, 6 inch diameter X 29 inch long, by placing the device on the slug and lifting it horizontal to the ground.

4.4.4 Cushion Bond (see dwgs 8865534 and 8880457). - The cushion bond on the handling device shall be tested using normal finger pressure around the edges of the handling device. The cushion shall not be grasped with the fingers to test the bond of the adhesive.

4.4.5 Latches (see dwgs 8865534 and 8880457). - The latch shall operate with normal pressure exerted with the thumbs.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. - The handling device shall be packaged in accordance with dwg 9206906 using the specified moisture barrier.

Source: https://assist.dla.mil -- Downloaded: 202640-18T04:36Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. - The handling device shall be packaged in accordance with dwg 9206906 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. - The handling device shall be packed in accordance with dwg 9206906.

5.3 Marking. - Marking shall be as specified on dwg 9206906.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.2 Objective evidence. - Records of contractor quality control and [illegible] inspections which can be verified.

6.3 Acceptance inspection equipment. - The contractor shall obtain approval of equipment designs, operating instructions, and calibration instructions from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1190-A008

Source: https://assist.dla.mil -- Downloaded: 202610-18T04:36Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-H-14705(MU) Handling Device Projectile: H4220 and H4224 (Training)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO. *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T04:36Z
Check the source to verify that this is the current version before use.